

DAV Quiz 1 - B

Roll No: _____

Name: _____

Instructions: Do not write on the question paper or tick any option. Write selected option inside the brackets given at the end of each question "Answer []"

1. The PCS framework evaluates the trustworthiness of data-driven results. What does the "C" stand for?

A) Consistency B) Correlation C) Computability D) Causality Answer []

2. The overarching goal of "Veridical" Data Science (Chapter 1) is to ensure that data science results are:

A) Truthful and reliably map back to real-world phenomena. B) As mathematically complex as possible. C) Solely focused on maximizing predictive accuracy, regardless of interpretability. D) Generated entirely without human intervention. Answer []

3. In the Data Science Life Cycle (Chapter 2), the phase immediately following Data Discovery/Acquisition is typically:

A) Explanatory Data Analysis B) Data Cleaning C) Advanced Modeling D) Formulating the initial question
Answer []

4. Which of the following is a key characteristic of the DSLC as described in the text?

A) It is highly iterative, often requiring data scientists to loop back to previous stages. B) It is a strictly linear process that never repeats. C) It entirely ignores the domain context in favor of algorithmic purity. D) It ends immediately once a dataset is cleaned. Answer []

5. According to Chapter 4 on data cleaning, what is a primary risk of manually editing a raw data file in a spreadsheet program like Excel?

A) Spreadsheets cannot handle numerical data. B) Manual edits are hard to track, making the cleaning process irreproducible. C) Spreadsheets automatically delete all missing values. D) It is illegal to use spreadsheets in data science. Answer []

6. Which of the following is an example of a common data cleaning task?

A) Formulating a domain question. B) Standardizing inconsistent date formats (e.g., converting "Jan 1, 2023" and "2023-01-01" to a single format). C) Building a random forest model. D) Deploying a dashboard for stakeholders. Answer []

7. Based on Chapter 5, what is the main difference between Exploratory Data Analysis (EDA) and Explanatory Analysis?

A) EDA uses tables, while Explanatory Analysis uses charts. B) EDA is for the data scientist to discover insights; Explanatory Analysis is for communicating specific findings to an audience. C) EDA requires machine learning; Explanatory Analysis only uses simple math. D) EDA is done at the very end of the project, while Explanatory Analysis is done at the beginning. Answer []

8. If you want to visualize the distribution of a single continuous numerical variable, which geometric object (geom) is most appropriate according to Chapter 5? Answer []

A) A pie chart B) A scatter plot C) A histogram D) A line plot of the data over time

9. What role does domain knowledge play across the Data Science Life Cycle?

A) It is only useful during the Question Formulation stage. B) It is irrelevant; data science is purely about algorithms. C) It is crucial at every stage, from guiding cleaning decisions to interpreting model results. D) It is only used to make the final report look more professional. Answer []

10. What are the 'three realms' that must be strongly connected to ensure trustworthy real-world decisions, as described in Chapter 1?

A) The real-world realm, the data realm, and the algorithmic realm. B) The business realm, the academic realm, and the technology realm. C) The prediction realm, the inference realm, and the description realm. D) The hardware realm, the software realm, and the deployment realm. Answer []

Quiz B (Section B) Answers:

1. C (Computability)
2. A (Truthful and reliably map back to real-world phenomena)
3. B (Data Cleaning)
4. A (It is highly iterative, often requiring data scientists to loop back)
5. B (Manual edits are hard to track, making the cleaning process irreproducible)
6. B (Standardizing inconsistent date formats)
7. B (EDA is for the data scientist... Explanatory Analysis is for communicating...)
8. C (A histogram)
9. C (It is crucial at every stage)
10. A (The real-world realm, the data realm, and the algorithmic realm.)